Supplementary Information

Semiempirical exploration of Eu(III) coordination in microsolvated homogalacturonan fragments: effects of the methylesterification pattern

S1. Computational details

i) Program and Hamiltonians

MOPAC 23.2.5 (open-source release). Sparkle/PM7 keywords for a production optimization: `PM7 CHARGE=q AUX GNORM=1.0 CYCLES=3000 THREADS=2 GEO-OK T=3D DUMP=30M BONDS EF LET EXTERNAL=eu_fix.txt`; RM1 lanthanide model: `RM1 CHARGE=q AUX GNORM=1.0 CYCLES=3000 THREADS=2 GEO-OK T=3D DUMP=30M BONDS EF LET`. Most runs that did not reach self-consistence from their starting geometry (22 RM1 runs and one Sparkle/PM7* run) were repeated from the same geometry with `SHIFT=15 PULAY CAMP ITRY=999` added to these keywords; 18 of the RM1 runs then converged, four failed again and the Sparkle/PM7* run exhausted the cycle budget.

ii) Sparkle/PM7* correction

Two lines, read through the EXTERNAL keyword: `XFAC_Eu O 2.207500` and `ALPB_Eu O 2.500000` (program defaults 0.1782 and 1.2286). Only the Eu–O diatomic core–core term is affected; all other Sparkle/PM7 parameters are unchanged. The pair was chosen on the rigid Eu–OH_2 scan so that the repulsive wall places the minimum at 2.4–2.5 Å; the five other scans, the aqua clusters and the monomer controls were computed afterwards without further adjustment (main text, Section 2.3).

iii) Coordinates and constraints

Inputs use mixed internal/Cartesian coordinates: every heavy atom and every hydrogen not bonded to oxygen is given in Cartesian coordinates with all three flags set to 1 (optimized); every hydrogen bonded to an oxygen (water, hydroxyl, acid O–H) is given in internal coordinates with its O–H bond length fixed (flag 0; 0.96 Å for water, and the MMFF94 value of the relaxed chain, 0.97–0.98 Å, for hydroxyl and acid groups) and its angle and dihedral optimized (flags 1). Example lines (water, hydroxyl): `H  0.96000 0  127.7500 1  14.0000 1  78  77  82` and `H  0.97550 0  106.5892 1  57.9615 1  1  2  3`. Nothing else is constrained.

iv) Starting structures

Chains assembled from the $^4C_1$ α-D-galacturonate residue with explicit glycosidic torsions (h3: φ = 75°, ψ = 105°; h2: 105°, 135°; cl: (105,165), (75,105), (90,165), (75,120), (105,135)) and relaxed with MMFF94

(distance-dependent dielectric, ε = 4r, torsions fixed). Eu placement by minimization of a penalty function on the target Eu–O distances (2.48 Å) with a soft clash term (heavy atoms ≥ 2.9 Å from Eu unless targeted); nine waters placed on the vertices of a tricapped trigonal prism around Eu (first shell, 2.48 Å) and, where a vertex is blocked by the chain, at 4.6 Å (second shell), followed by a random rotational search of each water for the best hydrogen-bonding orientation; in addition, two waters hydrogen-bonded to each free carboxylate not bound to Eu (main series and galacturonate controls; none or four in the hydration-sensitivity series, Table S5), so that a main-series cluster carries 9–21 waters.

S2. Descriptor definitions and quality control

Oxygen atoms of every final geometry are classified from connectivity (bond criterion 1.2 × sum of covalent radii): carboxylate (C6 carbon bonded to two oxygens neither of which carries H or a methyl carbon), carbonyl_acid / acid_OH (protonated acid), carbonyl_ester / ether_ester (methyl ester), hydroxyl (C–O–H), ring_or_glycosidic (C–O–C), water (O with two H and no C), hydroxide / oxonium / oxide (quality-control classes; none occurred in the converged set). The first coordination sphere is the set of oxygens within 3.0 Å of Eu. Residue-level descriptors use the atom map of each residue: a free carboxylate (kind A) is bidentate when both O6 atoms are within 3.0 Å and monodentate when one is; contacts through the O6 atoms of ester (kind M) or acid (kind H) residues are counted separately (`n_esters_bound`, `n_acids_bound`) and never as carboxylate contacts. `multi-COO` in Table 4 of the main text counts structures with ≥ 2 distinct kind-A residues coordinating Eu. Exclusion rule: CN < 6 marks a structure as physically inadequate (one case, PM7*_M6_cl_ester, CN = 4).

Columns of Table S1 / results_final_v2.csv: job; model (PM7 = Sparkle/PM7*, RM1); series (main, sens_hydration, control); pname / pattern; conf (h3, h2, cl); mode (bid, mono2, ester, hyd; controls: w8–w12, bid, hyd, ester); charge; status (converged = gradient norm < 1.0; cycles = cycle budget exhausted; ended = SCF failure); heat (kcal mol^{-1}, comparable only within identical composition); gnorm; ncycles; CN; n_pectin; n_water; f_pectin; f_water; n_<type> (composition); d_pectin_mean, d_water_mean, d_carbox_mean (Å); n_pectin_3.6 / n_water_3.6 (second-sphere window); n_water_defects; n_acidOH; n_residues_bound; residues_bound; n_carboxylates_bound; n_bidentate_carboxylate; n_monodentate_carboxylate; n_esters_bound; n_acids_bound; n_backbone_O; n_hydroxyl_O; motif; chain_rmsd (Å); torsions_final (φ/ψ per linkage); max_dtorsion (°); rerun (1 = from the repetition with SCF convergers, S1 i).

Table S1. All optimizations

Table S1. First-sphere descriptors of all 180 optimizations (cutoff 3.0 Å). Status: conv. = converged (gradient norm < 1.0 kcal mol^{-1} Å^{-1}); cycles = cycle budget exhausted (last geometry analysed); SCF = self-consistence not achieved. Motif notation as in the main text (bi / mono: bidentate / monodentate free carboxylate; est: ester oxygen; acid: acid oxygen; +O*n*: backbone or hydroxyl oxygens).

Job	Model	Pattern	Conf	Start	Status	CN	Pectin O	Water O	f_{pectin}	Eu–O(carb) Å	Motif
PM7_A6_h3_bid	PM7*	A6	h3	bid	conv.	9	2	7	0.222	2.466	A2:bi
PM7_A6_h3_mono2	PM7*	A6	h3	mono2	cycles	7	2	5	0.286	2.341	A4:mono+O5
PM7_A6_h3_hyd	PM7*	A6	h3	hyd	cycles	9	0	9	0.000	—	—
PM7_A6_h2_bid	PM7*	A6	h2	bid	conv.	9	2	7	0.222	2.471	A2:bi
PM7_A6_h2_mono2	PM7*	A6	h2	mono2	conv.	8	1	7	0.125	2.315	A1:mono
PM7_A6_h2_hyd	PM7*	A6	h2	hyd	cycles	9	0	9	0.000	—	—
PM7_A6_cl_bid	PM7*	A6	cl	bid	conv.	8	2	6	0.250	2.473	A2:bi
PM7_A6_cl_mono2	PM7*	A6	cl	mono2	conv.	7	1	6	0.143	2.229	A2:mono
PM7_A6_cl_hyd	PM7*	A6	cl	hyd	cycles	9	0	9	0.000	—	—
PM7_B3_h3_bid	PM7*	B3	h3	bid	conv.	8	2	6	0.250	2.461	A2:bi
RM1_B3_h3_bid	RM1	B3	h3	bid	conv.	9	3	6	0.333	2.489	A1:+O2 A2:bi
PM7_B3_h3_mono2	PM7*	B3	h3	mono2	conv.	7	1	6	0.143	2.258	A2:mono
RM1_B3_h3_mono2	RM1	B3	h3	mono2	conv.	9	3	6	0.333	2.310	A1:mono+O5 A2:mono
PM7_B3_h3_hyd	PM7*	B3	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_B3_h3_hyd	RM1	B3	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_B3_h2_bid	PM7*	B3	h2	bid	conv.	9	2	7	0.222	2.328	A2:bi
RM1_B3_h2_bid	RM1	B3	h2	bid	conv.	9	2	7	0.222	2.553	A2:bi
PM7_B3_h2_mono2	PM7*	B3	h2	mono2	conv.	8	3	5	0.375	2.351	A0:+O1 A1:mono+O5 O4
RM1_B3_h2_mono2	RM1	B3	h2	mono2	conv.	10	5	5	0.500	2.367	A0:mono+O5 O1 A1:mono+O5 O4
PM7_B3_h2_hyd	PM7*	B3	h2	hyd	cycles	9	0	9	0.000	—	—
RM1_B3_h2_hyd	RM1	B3	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_B3_cl_bid	PM7*	B3	cl	bid	conv.	7	2	5	0.286	2.471	A2:bi
RM1_B3_cl_bid	RM1	B3	cl	bid	conv.	9	4	5	0.444	2.426	A0:mono A1:+O2 A2:bi
PM7_B3_cl_mono2	PM7*	B3	cl	mono2	conv.	7	1	6	0.143	2.294	A2:mono
RM1_B3_cl_mono2	RM1	B3	cl	mono2	conv.	9	3	6	0.333	2.298	A1:mono+O5 A2:mono
PM7_B3_cl_hyd	PM7*	B3	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_B3_cl_hyd	RM1	B3	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_L3_h3_bid	PM7*	L3	h3	bid	conv.	9	2	7	0.222	2.482	A2:bi
RM1_L3_h3_bid	RM1	L3	h3	bid	conv.	9	2	7	0.222	2.543	A2:bi
PM7_L3_h3_mono2	PM7*	L3	h3	mono2	conv.	7	1	6	0.143	2.250	A2:mono
RM1_L3_h3_mono2	RM1	L3	h3	mono2	conv.	10	4	6	0.400	2.335	M1:+O1O2 A2:mono+O5 O4

Job	Model	Pattern	Conf	Start	Status	CN	Pectin O	Water O	f_{pectin}	Eu–O(carb) Å	Motif
PM7_L3_h3_hyd	PM7*	L3	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_L3_h3_hyd	RM1	L3	h3	hyd	SCF	9	0	9	0.000	—	—
PM7_L3_h2_bid	PM7*	L3	h2	bid	conv.	9	2	7	0.222	2.483	A2:bi
RM1_L3_h2_bid	RM1	L3	h2	bid	conv.	9	2	7	0.222	2.493	A2:bi
PM7_L3_h2_mono2	PM7*	L3	h2	mono2	conv.	8	2	6	0.250	2.310	M1:+O2 A2:mono
RM1_L3_h2_mono2	RM1	L3	h2	mono2	conv.	9	3	6	0.333	2.289	M1:+O2O3 A2:mono
PM7_L3_h2_hyd	PM7*	L3	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_L3_h2_hyd	RM1	L3	h2	hyd	SCF	9	0	9	0.000	—	—
PM7_L3_cl_bid	PM7*	L3	cl	bid	conv.	8	2	6	0.250	2.474	A2:bi
RM1_L3_cl_bid	RM1	L3	cl	bid	conv.	8	2	6	0.250	2.473	A2:bi
PM7_L3_cl_mono2	PM7*	L3	cl	mono2	conv.	7	1	6	0.143	2.322	A2:mono
RM1_L3_cl_mono2	RM1	L3	cl	mono2	conv.	8	2	6	0.250	2.337	M1:+O2 A2:mono
PM7_L3_cl_hyd	PM7*	L3	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_L3_cl_hyd	RM1	L3	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_P3_h3_bid	PM7*	P3	h3	bid	conv.	8	2	6	0.250	2.456	A1:bi
PM7_P3_h3_mono2	PM7*	P3	h3	mono2	conv.	7	1	6	0.143	2.267	A1:mono
PM7_P3_h3_hyd	PM7*	P3	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_P3_h2_bid	PM7*	P3	h2	bid	cycles	9	2	7	0.222	2.477	A1:bi
PM7_P3_h2_mono2	PM7*	P3	h2	mono2	conv.	9	1	8	0.111	2.316	A4:mono
PM7_P3_h2_hyd	PM7*	P3	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_P3_cl_bid	PM7*	P3	cl	bid	conv.	9	2	7	0.222	2.459	A1:bi
PM7_P3_cl_mono2	PM7*	P3	cl	mono2	conv.	8	3	5	0.375	2.368	A0:+O1 A1:mono+O5 O4
PM7_P3_cl_hyd	PM7*	P3	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_B2_h3_bid	PM7*	B2	h3	bid	conv.	8	2	6	0.250	2.467	A1:bi
RM1_B2_h3_bid	RM1	B2	h3	bid	conv.	9	3	6	0.333	2.473	A0:+O2 A1:bi
PM7_B2_h3_mono2	PM7*	B2	h3	mono2	conv.	7	1	6	0.143	2.258	A1:mono
RM1_B2_h3_mono2	RM1	B2	h3	mono2	conv.	9	3	6	0.333	2.286	A0:mono+O5 A1:mono
PM7_B2_h3_hyd	PM7*	B2	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_B2_h3_hyd	RM1	B2	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_B2_h2_bid	PM7*	B2	h2	bid	conv.	9	2	7	0.222	2.467	A1:bi
RM1_B2_h2_bid	RM1	B2	h2	bid	conv.	9	2	7	0.222	2.556	A1:bi
PM7_B2_h2_mono2	PM7*	B2	h2	mono2	cycles	8	3	5	0.375	2.352	A0:+O1 A1:mono+O5 O4
RM1_B2_h2_mono2	RM1	B2	h2	mono2	conv.	8	3	5	0.375	2.317	A0:+O1 A1:mono+O5 O4
PM7_B2_h2_hyd	PM7*	B2	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_B2_h2_hyd	RM1	B2	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_B2_cl_bid	PM7*	B2	cl	bid	conv.	8	2	6	0.250	2.434	A1:bi

Job	Model	Pattern	Conf	Start	Status	CN	Pectin O	Water O	f_{pectin}	Eu–O(carb) Å	Motif
RM1_B2_cl_bid	RM1	B2	cl	bid	conv.	8	2	6	0.250	2.575	A1:bi
PM7_B2_cl_mono2	PM7*	B2	cl	mono2	conv.	8	3	5	0.375	2.314	A0:+O1 A1:mono+O5 O4
RM1_B2_cl_mono2	RM1	B2	cl	mono2	conv.	9	4	5	0.444	2.316	A0:+O1O2 A1:mono+O5 O4
PM7_B2_cl_hyd	PM7*	B2	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_B2_cl_hyd	RM1	B2	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_D2_h3_bid	PM7*	D2	h3	bid	conv.	8	2	6	0.250	2.473	A3:bi
RM1_D2_h3_bid	RM1	D2	h3	bid	conv.	8	2	6	0.250	2.533	A3:bi
PM7_D2_h3_mono2	PM7*	D2	h3	mono2	conv.	9	2	7	0.222	2.474	A3:bi
RM1_D2_h3_mono2	RM1	D2	h3	mono2	conv.	9	2	7	0.222	2.560	A3:bi
PM7_D2_h3_hyd	PM7*	D2	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_D2_h3_hyd	RM1	D2	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_D2_h2_bid	PM7*	D2	h2	bid	conv.	9	2	7	0.222	2.479	A3:bi
RM1_D2_h2_bid	RM1	D2	h2	bid	SCF	9	2	7	0.222	2.503	A3:bi
PM7_D2_h2_mono2	PM7*	D2	h2	mono2	conv.	8	1	7	0.125	2.338	A3:mono
RM1_D2_h2_mono2	RM1	D2	h2	mono2	conv.	8	1	7	0.125	2.320	A3:mono
PM7_D2_h2_hyd	PM7*	D2	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_D2_h2_hyd	RM1	D2	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_D2_cl_bid	PM7*	D2	cl	bid	conv.	9	2	7	0.222	2.476	A3:bi
RM1_D2_cl_bid	RM1	D2	cl	bid	SCF	9	2	7	0.222	2.537	A3:bi
PM7_D2_cl_mono2	PM7*	D2	cl	mono2	conv.	8	2	6	0.250	2.453	A3:bi
RM1_D2_cl_mono2	RM1	D2	cl	mono2	SCF	8	2	6	0.250	2.565	A3:bi
PM7_D2_cl_hyd	PM7*	D2	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_D2_cl_hyd	RM1	D2	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_S1_h3_bid	PM7*	S1	h3	bid	conv.	9	2	7	0.222	2.469	A2:bi
RM1_S1_h3_bid	RM1	S1	h3	bid	conv.	9	2	7	0.222	2.471	A2:bi
PM7_S1_h3_hyd	PM7*	S1	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_S1_h3_hyd	RM1	S1	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_S1_h2_bid	PM7*	S1	h2	bid	conv.	9	2	7	0.222	2.471	A2:bi
RM1_S1_h2_bid	RM1	S1	h2	bid	conv.	9	2	7	0.222	2.471	A2:bi
PM7_S1_h2_hyd	PM7*	S1	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_S1_h2_hyd	RM1	S1	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_S1_cl_bid	PM7*	S1	cl	bid	conv.	8	2	6	0.250	2.466	A2:bi
RM1_S1_cl_bid	RM1	S1	cl	bid	conv.	8	2	6	0.250	2.455	A2:bi
PM7_S1_cl_hyd	PM7*	S1	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_S1_cl_hyd	RM1	S1	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_M6_h3_ester	PM7*	M6	h3	ester	cycles	7	2	5	0.286	—	M2:+O5 M3:est
RM1_M6_h3_ester	RM1	M6	h3	ester	conv.	8	3	5	0.375	—	M2:est+O5 M3:est
PM7_M6_h3_hyd	PM7*	M6	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_M6_h3_hyd	RM1	M6	h3	hyd	conv.	9	0	9	0.000	—	—

Job	Model	Pattern	Conf	Start	Status	CN	Pectin O	Water O	f_{pectin}	Eu–O(carb) Å	Motif
PM7_M6_h2_ester	PM7*	M6	h2	ester	cycles	4	1	3	0.250	—	M2:+O1 M3:+O4
RM1_M6_h2_ester	RM1	M6	h2	ester	conv.	9	4	5	0.444	—	M2:+O1O2 M3:est+O5O4
PM7_M6_h2_hyd	PM7*	M6	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_M6_h2_hyd	RM1	M6	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_M6_cl_ester	PM7*	M6	cl	ester	conv.	4	1	3	0.250	—	M2:+O1 M3:+O4
RM1_M6_cl_ester	RM1	M6	cl	ester	conv.	8	3	5	0.375	—	M2:+O1 M3:est+O5O4
PM7_M6_cl_hyd	PM7*	M6	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_M6_cl_hyd	RM1	M6	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_H3alt_h3_bid	PM7*	H3alt	h3	bid	conv.	8	2	6	0.250	2.466	A2:bi
RM1_H3alt_h3_bid	RM1	H3alt	h3	bid	conv.	8	2	6	0.250	2.537	A2:bi
PM7_H3alt_h3_mono2	PM7*	H3alt	h3	mono2	conv.	8	1	7	0.125	2.334	A2:mono
RM1_H3alt_h3_mono2	RM1	H3alt	h3	mono2	conv.	8	1	7	0.125	2.280	A2:mono
PM7_H3alt_h3_hyd	PM7*	H3alt	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_H3alt_h3_hyd	RM1	H3alt	h3	hyd	SCF	9	0	9	0.000	—	—
PM7_H3alt_h2_bid	PM7*	H3alt	h2	bid	conv.	9	2	7	0.222	2.480	A2:bi
RM1_H3alt_h2_bid	RM1	H3alt	h2	bid	conv.	9	2	7	0.222	2.490	A2:bi
PM7_H3alt_h2_mono2	PM7*	H3alt	h2	mono2	conv.	8	2	6	0.250	2.283	H1:+O2 A2:mono
RM1_H3alt_h2_mono2	RM1	H3alt	h2	mono2	conv.	9	3	6	0.333	2.282	H1:+O2O3 A2:mono
PM7_H3alt_h2_hyd	PM7*	H3alt	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_H3alt_h2_hyd	RM1	H3alt	h2	hyd	conv.	9	0	9	0.000	—	—
PM7_H3alt_cl_bid	PM7*	H3alt	cl	bid	conv.	7	2	5	0.286	2.460	A2:bi
RM1_H3alt_cl_bid	RM1	H3alt	cl	bid	conv.	8	3	5	0.375	2.461	H1:+O2 A2:bi
PM7_H3alt_cl_mono2	PM7*	H3alt	cl	mono2	conv.	8	2	6	0.250	2.290	H1:+O2 A2:mono
RM1_H3alt_cl_mono2	RM1	H3alt	cl	mono2	conv.	9	3	6	0.333	2.480	H1:+O2 A2:bi
PM7_H3alt_cl_hyd	PM7*	H3alt	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_H3alt_cl_hyd	RM1	H3alt	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_H3blk_h3_bid	PM7*	H3blk	h3	bid	conv.	8	2	6	0.250	2.462	A2:bi
RM1_H3blk_h3_bid	RM1	H3blk	h3	bid	conv.	8	2	6	0.250	2.585	A2:bi
PM7_H3blk_h3_mono2	PM7*	H3blk	h3	mono2	conv.	7	1	6	0.143	2.236	A2:mono
RM1_H3blk_h3_mono2	RM1	H3blk	h3	mono2	conv.	9	3	6	0.333	2.300	A1:mono+O5 A2:mono
PM7_H3blk_h3_hyd	PM7*	H3blk	h3	hyd	conv.	9	0	9	0.000	—	—
RM1_H3blk_h3_hyd	RM1	H3blk	h3	hyd	conv.	9	0	9	0.000	—	—
PM7_H3blk_h2_bid	PM7*	H3blk	h2	bid	conv.	9	2	7	0.222	2.460	A2:bi
RM1_H3blk_h2_bid	RM1	H3blk	h2	bid	conv.	9	2	7	0.222	2.547	A2:bi
PM7_H3blk_h2_mono2	PM7*	H3blk	h2	mono2	conv.	8	1	7	0.125	2.289	A1:mono
RM1_H3blk_h2_mono2	RM1	H3blk	h2	mono2	conv.	8	1	7	0.125	2.273	A1:mono
PM7_H3blk_h2_hyd	PM7*	H3blk	h2	hyd	conv.	9	0	9	0.000	—	—
RM1_H3blk_h2_hyd	RM1	H3blk	h2	hyd	conv.	9	0	9	0.000	—	—

Job	Model	Pattern	Conf	Start	Status	CN	Pectin O	Water O	f_{pectin}	Eu–O(carb) Å	Motif
PM7_H3blk_cl_bid	PM7*	H3blk	cl	bid	conv.	9	2	7	0.222	2.468	A2:bi
RM1_H3blk_cl_bid	RM1	H3blk	cl	bid	conv.	9	2	7	0.222	2.557	A2:bi
PM7_H3blk_cl_mono2	PM7*	H3blk	cl	mono2	conv.	7	1	6	0.143	2.221	A2:mono
RM1_H3blk_cl_mono2	RM1	H3blk	cl	mono2	conv.	9	3	6	0.333	2.302	A1:mono+O5 A2:mono
PM7_H3blk_cl_hyd	PM7*	H3blk	cl	hyd	conv.	9	0	9	0.000	—	—
RM1_H3blk_cl_hyd	RM1	H3blk	cl	hyd	conv.	9	0	9	0.000	—	—
PM7_A6_h2_bid_cw0	PM7*	A6	h2	bid	conv.	9	2	7	0.222	2.476	A2:bi
PM7_A6_h2_bid_cw4	PM7*	A6	h2	bid	cycles	9	2	7	0.222	2.469	A2:bi
PM7_B3_h2_bid_cw0	PM7*	B3	h2	bid	conv.	9	2	7	0.222	2.462	A2:bi
RM1_B3_h2_bid_cw0	RM1	B3	h2	bid	conv.	9	2	7	0.222	2.552	A2:bi
PM7_B3_h2_bid_cw4	PM7*	B3	h2	bid	cycles	9	2	7	0.222	2.325	A2:bi
RM1_B3_h2_bid_cw4	RM1	B3	h2	bid	conv.	9	2	7	0.222	2.550	A2:bi
PM7_L3_h2_bid_cw0	PM7*	L3	h2	bid	conv.	9	2	7	0.222	2.481	A2:bi
RM1_L3_h2_bid_cw0	RM1	L3	h2	bid	conv.	9	2	7	0.222	2.491	A2:bi
PM7_L3_h2_bid_cw4	PM7*	L3	h2	bid	conv.	9	2	7	0.222	2.484	A2:bi
RM1_L3_h2_bid_cw4	RM1	L3	h2	bid	conv.	9	2	7	0.222	2.499	A2:bi
PM7_P3_h2_bid_cw0	PM7*	P3	h2	bid	conv.	9	2	7	0.222	2.484	A1:bi
PM7_P3_h2_bid_cw4	PM7*	P3	h2	bid	conv.	9	2	7	0.222	2.486	A1:bi
PM7_CTRL_Eu_w8	PM7*	CTRL_water	-	w8	conv.	8	0	8	0.000	—	—
PM7_CTRL_Eu_w9	PM7*	CTRL_water	-	w9	conv.	9	0	9	0.000	—	—
PM7_CTRL_Eu_w10	PM7*	CTRL_water	-	w10	conv.	9	0	9	0.000	—	—
PM7_CTRL_Eu_w12	PM7*	CTRL_water	-	w12	conv.	9	0	9	0.000	—	—
PM7_CTRL_GalA_bid	PM7*	CTRL_GalA	-	bid	conv.	9	2	7	0.222	2.463	A0:bi
PM7_CTRL_GalA_hyd	PM7*	CTRL_GalA	-	hyd	conv.	9	0	9	0.000	—	—
PM7_CTRL_GalAMe_ester	PM7*	CTRL_GalAMe	-	ester	conv.	9	1	8	0.111	—	M0:est
PM7_CTRL_GalAMe_hyd	PM7*	CTRL_GalAMe	-	hyd	conv.	9	0	9	0.000	—	—
PM7_CTRL_acetate_bid	PM7*	CTRL_acetate	-	bid	conv.	9	2	7	0.222	2.458	—
RM1_CTRL_Eu_w8	RM1	CTRL_water	-	w8	conv.	8	0	8	0.000	—	—
RM1_CTRL_Eu_w9	RM1	CTRL_water	-	w9	conv.	9	0	9	0.000	—	—
RM1_CTRL_Eu_w10	RM1	CTRL_water	-	w10	conv.	9	0	9	0.000	—	—
RM1_CTRL_Eu_w12	RM1	CTRL_water	-	w12	conv.	9	0	9	0.000	—	—
RM1_CTRL_GalA_bid	RM1	CTRL_GalA	-	bid	conv.	9	2	7	0.222	2.450	A0:bi
RM1_CTRL_GalA_hyd	RM1	CTRL_GalA	-	hyd	conv.	9	0	9	0.000	—	—
RM1_CTRL_GalAMe_ester	RM1	CTRL_GalAMe	-	ester	conv.	9	1	8	0.111	—	M0:est
RM1_CTRL_GalAMe_hyd	RM1	CTRL_GalAMe	-	hyd	conv.	9	0	9	0.000	—	—
RM1_CTRL_acetate_bid	RM1	CTRL_acetate	-	bid	conv.	9	2	7	0.222	2.443	—

Table S2. The 17 optimizations that did not reach the convergence criterion (excluded from all statistics). Descriptors refer to the last available geometry.

Job	Model	Reason	Cycles	CN	f_{pectin}	Motif (last geometry)
PM7_A6_h3_mono2	PM7*	cycle budget exhausted	3000	7	0.286	A4:mono+O5
PM7_A6_h3_hyd	PM7*	cycle budget exhausted	3000	9	0.000	—
PM7_A6_h2_hyd	PM7*	cycle budget exhausted	3000	9	0.000	—
PM7_A6_cl_hyd	PM7*	cycle budget exhausted	3000	9	0.000	—
PM7_B3_h2_hyd	PM7*	cycle budget exhausted	3000	9	0.000	—
RM1_L3_h3_hyd	RM1	SCF not converged	9	9	0.000	—
RM1_L3_h2_hyd	RM1	SCF not converged	7	9	0.000	—
PM7_P3_h2_bid	PM7*	cycle budget exhausted	3000	9	0.222	A1:bi
PM7_B2_h2_mono2	PM7*	cycle budget exhausted	3000	8	0.375	A0:+O1 A1:mono+O5O4
RM1_D2_h2_bid	RM1	SCF not converged	2	9	0.222	A3:bi
RM1_D2_cl_bid	RM1	SCF not converged	4	9	0.222	A3:bi
RM1_D2_cl_mono2	RM1	SCF not converged	3	8	0.250	A3:bi
PM7_M6_h3_ester	PM7*	cycle budget exhausted	3000	7	0.286	M2:+O5 M3:est
PM7_M6_h2_ester	PM7*	cycle budget exhausted	3000	4	0.250	M2:+O1 M3:+O4
RM1_H3alt_h3_hyd	RM1	SCF not converged	5	9	0.000	—
PM7_A6_h2_bid_cw4	PM7*	cycle budget exhausted	3000	9	0.222	A2:bi
PM7_B3_h2_bid_cw4	PM7*	cycle budget exhausted	3000	9	0.222	A2:bi

Table S3. Rigid Eu–O scans

Table S3a. Sparkle/PM7: binding energy (kcal mol⁻¹) relative to Eu^{3+} and the rigid ligand as a function of the Eu–O distance (Å). Ligand geometries and approach directions are identical for the three Hamiltonians (ligands.py). Acetate is bidentate with Eu on the C_2 axis; ester = methyl acetate coordinated through the carbonyl oxygen (C=O···Eu = 140°); ether = dimethyl ether on the C–O–C bisector.

Eu–O (Å)	Water	Methanol	Ether	Ester	Acetate	Methoxide
1.2	147.4	162.7	185.4	65.4	619.9	-410.8
1.3	80.9	98.9	115.6	7.9	228.1	-444.2
1.4	43.2	64.7	77.8	-21.4	41.7	-450.9
1.5	18.7	44.3	58.9	-36.3	-85.1	-446.4
1.6	1.6	30.9	52.4	-42.2	-173.0	-437.2
1.7	-10.4	20.3	51.1	-43.5	-234.0	-427.4
1.8	-18.3	9.5	47.2	-44.9	-276.8	-419.9
1.9	-22.6	-2.6	35.5	-49.7	-306.6	-415.8
2.0	-24.2	-14.5	16.5	-57.6	-326.8	-413.3
2.1	-24.8	-24.2	-4.4	-65.6	-338.7	-409.9
2.2	-26.3	-31.0	-21.9	-71.2	-342.7	-404.1
2.3	-29.2	-35.4	-33.2	-73.1	-338.8	-395.5
2.4	-31.7	-37.4	-38.2	-71.5	-329.2	-384.2
2.5	-30.9	-35.8	-37.3	-66.0	-316.7	-369.8
2.6	-24.9	-29.3	-30.7	-56.0	-301.9	-351.6
2.7	-14.9	-19.2	-20.5	-43.1	-284.6	-331.0
2.8	-5.9	-10.1	-11.5	-32.0	-269.4	-312.5
2.9	-3.5	-7.7	-9.2	-27.8	-264.6	-301.4
3.0	-9.1	-13.3	-14.9	-31.8	-273.4	-298.8
3.2	-27.3	-31.2	-32.9	-47.2	-301.8	-301.7
3.4	-33.5	-37.1	-38.8	-50.9	-305.3	-294.1
3.6	-32.2	-35.6	-37.1	-47.6	-293.9	-280.3

Table S3b. Sparkle/PM7*: binding energy (kcal mol⁻¹) relative to Eu^{3+} and the rigid ligand as a function of the Eu–O distance (Å). Ligand geometries and approach directions are identical for the three Hamiltonians (ligands.py). Acetate is bidentate with Eu on the C_2 axis; ester = methyl acetate coordinated through the carbonyl oxygen (C=O···Eu = 140°); ether = dimethyl ether on the C–O–C bisector.

Eu–O (Å)	Water	Methanol	Ether	Ester	Acetate	Methoxide
1.2	759.9	775.1	797.9	673.9	1844.8	201.7
1.3	491.4	509.3	526.1	415.2	1049.0	-33.7
1.4	313.2	334.7	347.7	246.2	581.6	-180.9
1.5	191.3	216.9	231.5	134.6	260.2	-273.8
1.6	107.2	136.6	158.0	62.2	38.2	-331.6
1.7	49.5	80.2	111.0	15.5	-114.2	-367.5
1.8	10.8	38.6	76.4	-16.4	-218.5	-390.8
1.9	-13.8	6.2	44.3	-41.4	-289.0	-407.0
2.0	-28.4	-18.7	12.3	-62.1	-335.2	-417.5
2.1	-36.9	-36.4	-16.6	-78.0	-363.0	-422.1
2.2	-42.9	-47.6	-38.5	-87.9	-375.9	-420.7
2.3	-47.8	-54.1	-51.8	-91.9	-376.2	-414.1
2.4	-50.8	-56.6	-57.3	-90.7	-367.5	-403.4
2.5	-49.5	-54.4	-55.9	-84.6	-353.9	-388.4
2.6	-42.3	-46.7	-48.1	-73.4	-336.7	-369.0
2.7	-30.8	-35.0	-36.3	-59.0	-316.3	-346.9
2.8	-20.0	-24.3	-25.6	-46.1	-297.6	-326.6
2.9	-15.8	-20.1	-21.5	-40.1	-289.3	-313.7
3.0	-19.7	-23.9	-25.5	-42.4	-294.6	-309.4
3.2	-34.7	-38.7	-40.4	-54.6	-316.6	-309.2
3.4	-38.3	-42.0	-43.7	-55.8	-315.1	-299.0
3.6	-35.1	-38.6	-40.1	-50.5	-299.8	-283.3

Table S3c. RM1: binding energy (kcal mol^{-1}) relative to Eu^{3+} and the rigid ligand as a function of the Eu–O distance (Å). Ligand geometries and approach directions are identical for the three Hamiltonians (ligands.py). Acetate is bidentate with Eu on the C_2 axis; ester = methyl acetate coordinated through the carbonyl oxygen (C=O···Eu = 140°); ether = dimethyl ether on the C–O–C bisector.

Eu–O (Å)	Water	Methanol	Ether	Ester	Acetate	Methoxide
1.2	428.7	386.3	344.9	301.5	933.3	-180.6
1.3	302.1	264.1	225.6	184.1	499.9	-285.8
1.4	207.1	173.6	138.5	99.0	235.6	-359.5
1.5	137.1	107.7	75.7	38.3	59.1	-409.5
1.6	83.6	57.9	28.4	-6.9	-74.9	-444.7
1.7	39.0	16.4	-11.1	-44.3	-188.9	-473.0
1.8	-1.3	-21.0	-46.8	-78.0	-289.4	-498.6
1.9	-37.8	-54.9	-78.8	-108.3	-375.5	-521.8
2.0	-68.2	-83.1	-105.0	-133.0	-443.9	-540.3
2.1	-90.3	-103.4	-123.3	-149.7	-492.6	-551.9
2.2	-103.3	-115.3	-133.4	-158.2	-522.3	-555.8
2.3	-108.5	-119.9	-136.6	-159.6	-536.0	-553.2
2.4	-108.2	-119.2	-135.0	-156.0	-538.3	-545.8
2.5	-104.3	-115.0	-130.3	-149.3	-532.9	-535.7
2.6	-98.5	-108.8	-123.5	-140.9	-522.9	-524.2
2.7	-91.7	-101.5	-115.6	-131.6	-510.3	-512.1
2.8	-84.3	-93.6	-107.0	-121.9	-496.4	-500.0
2.9	-76.6	-85.5	-98.2	-112.2	-482.2	-488.3
3.0	-68.9	-77.5	-89.5	-102.9	-468.0	-477.6
3.2	-55.0	-63.2	-74.5	-86.9	-442.3	-464.8
3.4	-44.8	-53.2	-65.2	-76.0	-423.5	-459.7
3.6	-37.4	-46.9	-61.0	-69.6	-411.1	-458.7

Table S4. Control calculations

Table S4. Controls: $[Eu(H_2O)_n]^{3+}$ (n = 8, 9, 10, 12), galacturonate monomer (bidentate and hydrated starts), methyl galacturonate (ester and hydrated starts) and acetate. Heats of formation are comparable only between entries of identical composition (e.g. w9 − w8 = ninth-water binding energy).

Job	Model	System	Start	*q*	Status	CN	Eu–OH_2 mean (Å)	Eu–O(carb) (Å)	ΔH_f (kcal mol^{-1})
PM7_CTRL_Eu_w8	PM7*	CTRL_water	w8	3	converged	8	2.455	—	242.75
PM7_CTRL_Eu_w9	PM7*	CTRL_water	w9	3	converged	9	2.460	—	171.46
PM7_CTRL_Eu_w10	PM7*	CTRL_water	w10	3	converged	9	2.460	—	84.56
PM7_CTRL_Eu_w12	PM7*	CTRL_water	w12	3	converged	9	2.459	—	-88.59
PM7_CTRL_GalA_bid	PM7*	CTRL_GalA	bid	2	converged	9	2.459	2.463	-462.65
PM7_CTRL_GalA_hyd	PM7*	CTRL_GalA	hyd	2	converged	9	2.452	—	-605.68
PM7_CTRL_GalAMe_ester	PM7*	CTRL_GalAMe	ester	3	converged	9	2.459	—	-227.46
PM7_CTRL_GalAMe_hyd	PM7*	CTRL_GalAMe	hyd	3	converged	9	2.457	—	-218.33
PM7_CTRL_acetate_bid	PM7*	CTRL_acetate	bid	2	converged	9	2.458	2.458	-247.58
RM1_CTRL_Eu_w8	RM1	CTRL_water	w8	3	converged	8	2.428	—	204.29
RM1_CTRL_Eu_w9	RM1	CTRL_water	w9	3	converged	9	2.447	—	104.13
RM1_CTRL_Eu_w10	RM1	CTRL_water	w10	3	converged	9	2.445	—	24.25
RM1_CTRL_Eu_w12	RM1	CTRL_water	w12	3	converged	9	2.440	—	-129.81
RM1_CTRL_GalA_bid	RM1	CTRL_GalA	bid	2	converged	9	2.435	2.450	-555.82
RM1_CTRL_GalA_hyd	RM1	CTRL_GalA	hyd	2	converged	9	2.495	—	-764.58
RM1_CTRL_GalAMe_ester	RM1	CTRL_GalAMe	ester	3	converged	9	2.443	—	-274.25
RM1_CTRL_GalAMe_hyd	RM1	CTRL_GalAMe	hyd	3	converged	9	2.440	—	-267.15
RM1_CTRL_acetate_bid	RM1	CTRL_acetate	bid	2	converged	9	2.434	2.443	-337.37

Table S5. Hydration-sensitivity series (bidentate starts on the 2_1 helix): baselines without water on the free carboxylates and variants with four water molecules hydrogen-bonded to each free carboxylate not bound to the metal (main series: two).

Job	Model	Pattern	Variant	Status	CN	f_{pectin}	Eu–O(carb) (Å)	Motif
PM7_A6_h2_bid_cw0	PM7*	A6	0 H_2O per free carboxylate	converged	9	0.222	2.476	A2:bi
PM7_A6_h2_bid_cw4	PM7*	A6	4 H_2O per free carboxylate	-	9	0.222	2.469	A2:bi
PM7_B3_h2_bid_cw0	PM7*	B3	0 H_2O per free carboxylate	converged	9	0.222	2.462	A2:bi
RM1_B3_h2_bid_cw0	RM1	B3	0 H_2O per free carboxylate	converged	9	0.222	2.552	A2:bi
PM7_B3_h2_bid_cw4	PM7*	B3	4 H_2O per free carboxylate	-	9	0.222	2.325	A2:bi
RM1_B3_h2_bid_cw4	RM1	B3	4 H_2O per free carboxylate	converged	9	0.222	2.550	A2:bi
PM7_L3_h2_bid_cw0	PM7*	L3	0 H_2O per free carboxylate	converged	9	0.222	2.481	A2:bi
RM1_L3_h2_bid_cw0	RM1	L3	0 H_2O per free carboxylate	converged	9	0.222	2.491	A2:bi
PM7_L3_h2_bid_cw4	PM7*	L3	4 H_2O per free carboxylate	converged	9	0.222	2.484	A2:bi
RM1_L3_h2_bid_cw4	RM1	L3	4 H_2O per free carboxylate	converged	9	0.222	2.499	A2:bi
PM7_P3_h2_bid_cw0	PM7*	P3	0 H_2O per free carboxylate	converged	9	0.222	2.484	A1:bi
PM7_P3_h2_bid_cw4	PM7*	P3	4 H_2O per free carboxylate	converged	9	0.222	2.486	A1:bi

Supplementary figures

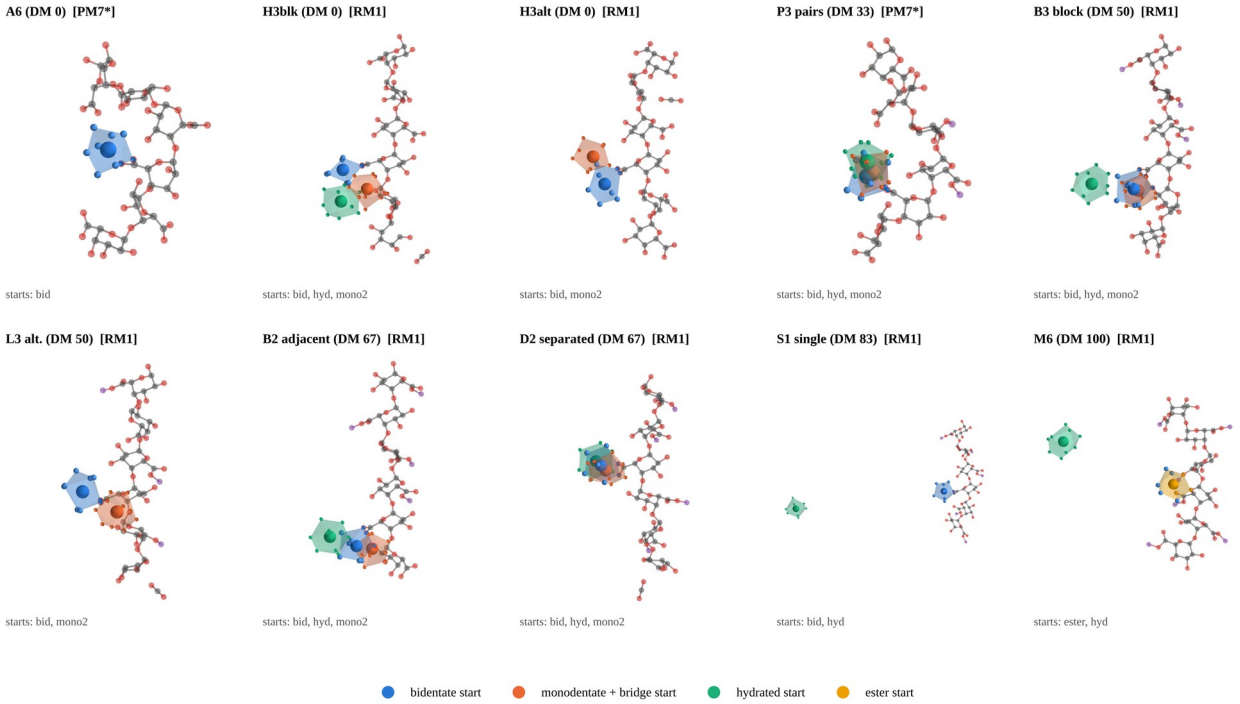


Figure S1. Converged structures from the different starting placements on the 3_1-like helix (h3), superposed on the chain (as Figure 8 of the main text; blue: bidentate start, orange: monodentate-plus-bridge start, green: hydrated start, gold: ester start).

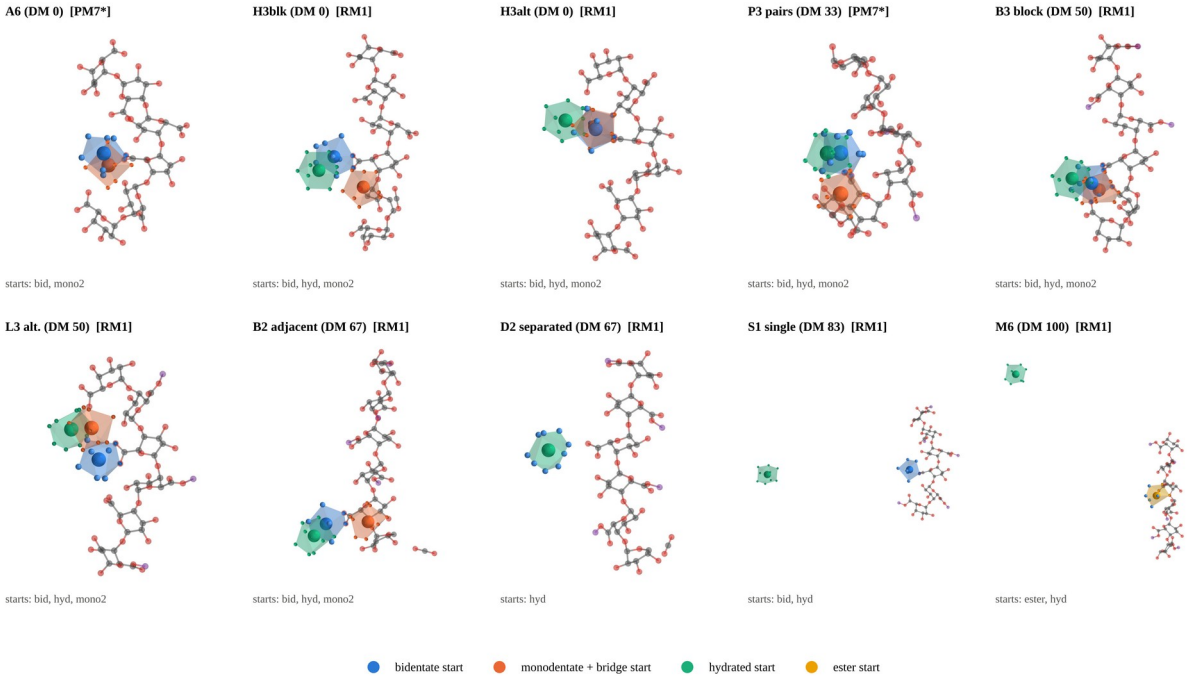


Figure S2. As Figure S1, for the irregular-coil conformer (cl).

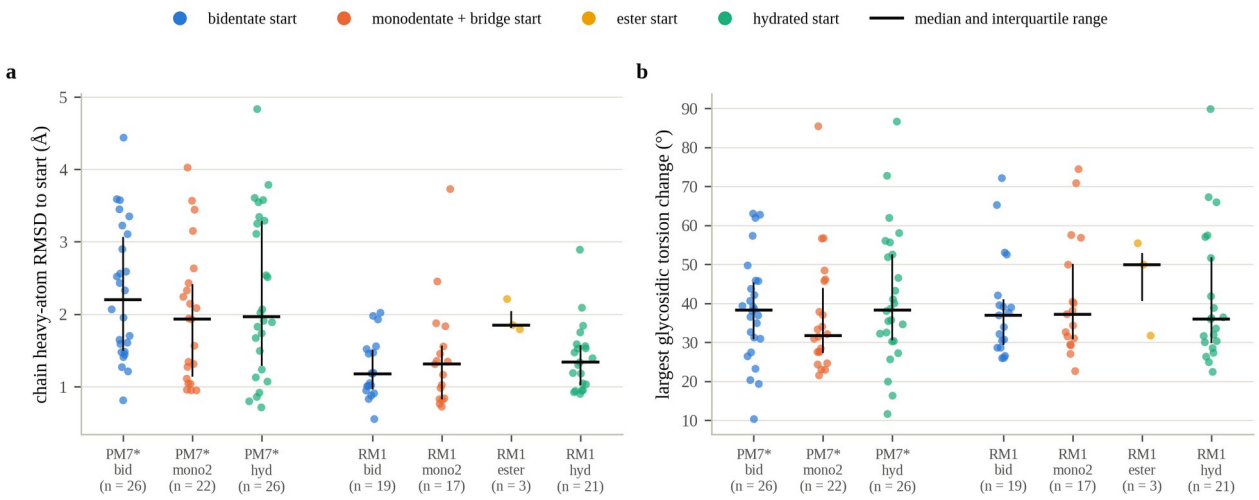


Figure S3. Chain deformation during optimization for converged main-series structures: (**a**) heavy-atom RMSD of the chain relative to its starting geometry and (**b**) largest change of any glycosidic torsion, by Hamiltonian and starting placement (bars: median and interquartile range). Hydrated starts, in which no pectin oxygen enters the first sphere, deform as much as the inner-sphere starts.